

Universal Extensions for CathodoLuminescence, Photoluminescence and Raman Spectroscopy

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Boost your Electron Microscope with Hyperspectral & Imaging Solutions



Clue Series

Add Cathodoluminescence and Raman Capabilities to your Electron Microscope

Nanomaterials

Optoelectronics
Semi-conductors

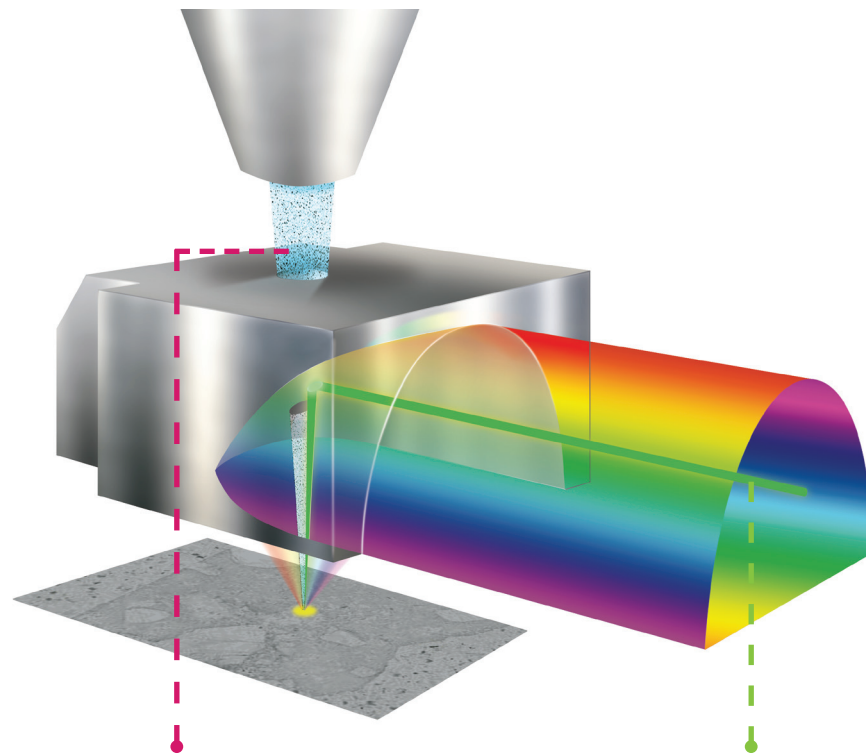
Geology
Mineralogy

Photovoltaics

HORIBA Scientific is your partner in spectroscopy with unequalled experience in optics, vacuum and detectors designs.

Based on combined experience in Japan (MP series) and in France (CLUE series), HORIBA Scientific offers a bunch of cathodoluminescence systems for process applications in semi-conductors as well as Research applications in Earth, Life and Material Sciences

With a consistent worldwide installed base and as a leader in diffraction gratings, Raman, PL and VUV spectroscopy, HORIBA Scientific is a global company with application and service support next to you.



What is SEM-CL Spectroscopy?

CathodoLuminescence (CL) is a non-destructive technique providing **maps of optical and electronic properties** of many kinds of materials with a nanometric spatial resolution.

Cathodoluminescence is similar to Photoluminescence (PL) technique, but the excitation by **high energy electron beam** can produce all the transitions to the higher energy excitation states and induce light emission from DUV to NIR.

CL technique is particularly suitable for analysis of particles, group IV semiconductors, thin films and nanostructures, novel photonics materials, oxides and minerals.

What is SEM-Raman Spectroscopy?

Raman spectroscopy is based upon the interaction of monochromatic light such like a **laser** with the molecules within a material.

This technique typically provides information about **chemical structure, phase and polymorphy, crystallinity, and molecular orientation**.

Raman analysis can be applied to any kind of organic (biological, polymers...) and inorganic (minerals, semi-conductors, glasses, oxides...) materials, except pure metals.

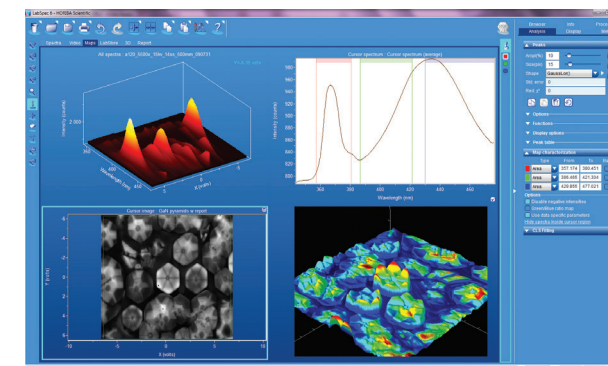
Gain unique insight into the chemical and electronic properties of materials with nanoscale resolution!

Measuring CL and Raman / PL photons inside the electron microscope* specimen chamber, in a **SameSpot™** configuration, permits to measure the following parameters in the same conditions of pressure and temperature as other EM measurements (EDX, XRF, EBSD, Auger...).

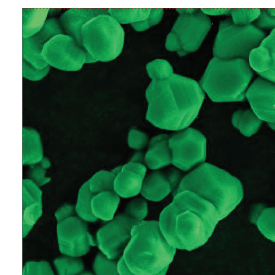
*Compatible with SEM and FIB/SEM microscopes

Cathodoluminescence

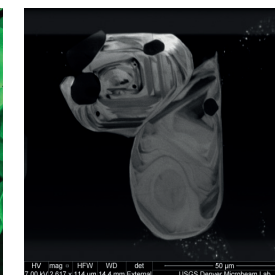
- Electrical and compositional properties
- Quality control and failure analysis
- Defects, impurities, dopants, vacancies
- Contaminations and inclusions
- Zoning analysis in zircons



GaN pyramidal nanostructures



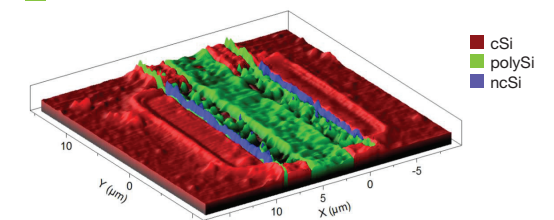
CL image of phosphor powder



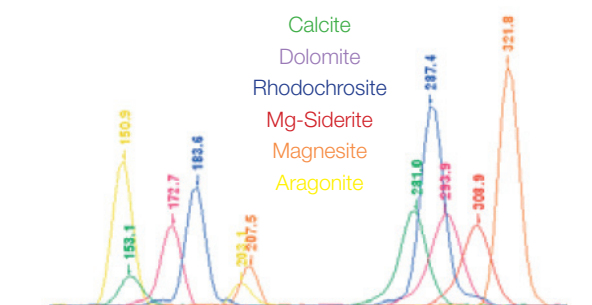
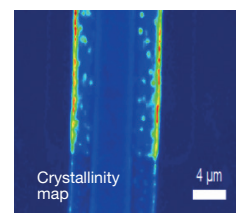
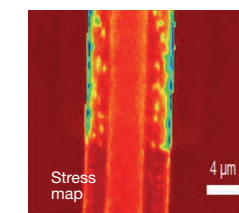
Zoning mosaic image of Zircons (400x400μm)

Raman Spectroscopy

- Compounds molecular identification
- Functional groups analysis
- Crystallinity, lattice modes
- Strain & Stress
- Free carrier concentration



2D and 3D Raman fast map of Si peak bandwidth, showing various crystallinity



Raman spectra of different types of carbonate crystals

CLUE Just Fits IN whatever your SEM!

Add-on Detectors Portfolio

An ultra-compact scalable solution to fit any request and budget

From i-CLUE affordable CL panchromatic imaging controlled by the electron microscope software, to F-CLUE fiber coupled ruggedized configurations and H-CLUE laboratory solutions featuring high performance direct coupling; HORIBA Scientific

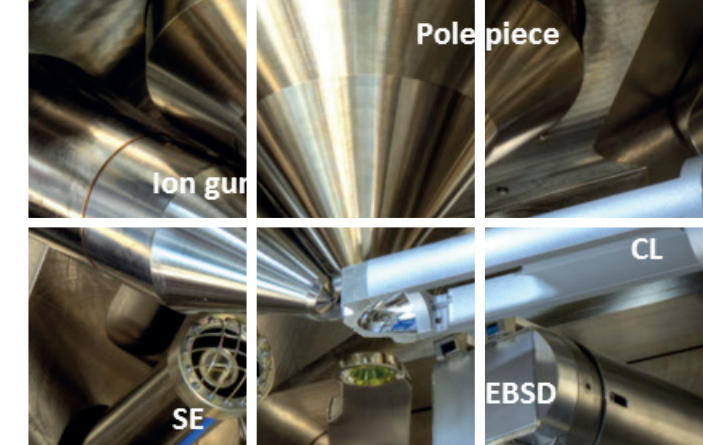
offers a cathodoluminescence solution to every customer and every sample.

In addition, our special product team can develop a custom solution to meet your specific requirements.

Keep the original configuration of your electron microscope

Because it is fully retractable, HORIBA CLUE series offers the possibility to add CL or Raman spectroscopy as a field upgrade on any famous brand of Electron Microscope*, without any restriction on microscope performance!

* Need a free horizontal port and technical compatibility validation



Affordable • Compact • Fast

i-CLUE

Fast Imaging CL

Best price-to-performance ratio for a plug-and-play fast CL imaging module features:

- Ultra-compact panchromatic CL detection
- Manually retractable mirror
- Large field of view ellipsoidal mirror collection
- Driven by EM software
- Imaging touch-screen controller



F-CLUE

Compact Hyperspectral CL

Flexible solution to fit any kind of environment, featuring:

- Imaging and/or hyperspectral CL
- Rugged optical fiber interface
- Selection of ellipsoidal or parabolic retractable mirror to fit all sample types and applications
- Powered by LabSpec 6™ Software
- Fully motorized spectrometer
- Optional RGB filters



H-CLUE

Versatile Hyperspectral CL

Free space optical coupling for the highest performance over the widest CL spectral range

- Fully automated control
- Motorized RPM with fine adjustment
- No chromatic shift
- Customized configurations: angle-resolved CL, time-resolved CL, polarization, filter wheels
- Imaging and spectroscopy from DUV to NIR with up to 5 detectors
- Long focal length spectrometers



R-CLUE

Raman PL & CL

All-in-one fully automated solution for high-end Research labs, featuring:

- Fiber coupled interface
- Colocalized laser and e-beam excitations
- Fully computer controlled
- Multi-wavelengths Raman lasers
- Injection-rejection Raman filters
- Field upgradeable
- Built-in video camera



Ultimate User Experience with LabSpec 6 Software

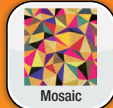


The core of the CLUE series is LabSpec 6™ software, concentrating the power of **advanced spectrum analysis, the best imaging and automation and unrivaled ease of use.**

LabSpec 6™ stands at the forefront of mosaic visualization, 2D & 3D hyperspectral imaging with built-in multivariate analysis, automated Particle analysis, multi-units (eV, nm, cm⁻¹...), spectral and intensity calibration.

3 Steps Workflow for Publication-Ready Results

Ease-of-Use



Start with AutoCal to ensure your system is spectrally and intensity calibrated; then insert your collection mirror in OneClick™ and navigate in your low and high magnification sample image with the unique EasyNav™ app, get your particle size and shape distribution and automate their spectral analysis.

Powerful Image Analysis



Easily identify present compounds by comparing unknown spectra with Raman spectral database, load spectra to create your multivariate analysis map and overlay the spectral image with SE image. Automate processing of multiple images and runaway!

Working Together



Because SEM is a multi-detector and multi-user instrument, our fully validated software allows to create multiple user accounts (administrator/operator), personalized hardware and software configurations. It includes as well templates for readily editable test reports.

The Best of Spectral Analysis For High Resolution Cathodoluminescence Images

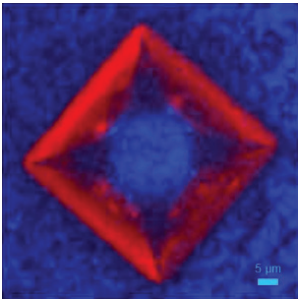
Unveil every detail of minerals and zircons with high resolution CL images. Map opto-electronic properties with nanometric scale resolution and reveal the true nature of ceramics, photonic nanomaterials, LED or photovoltaics thin-films, 2D heterostructures...



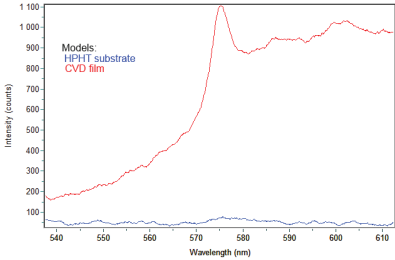
« H-CLUE unique achromatic design allows to measure easily CL signals even in the DUV, such like synthetic diamonds emissions. Its mirror is fully retractable; original configuration, performances and vacuum of SEM were not affected by the CL upgrade »

Dr A. Tallaire – CNRS - LSPM

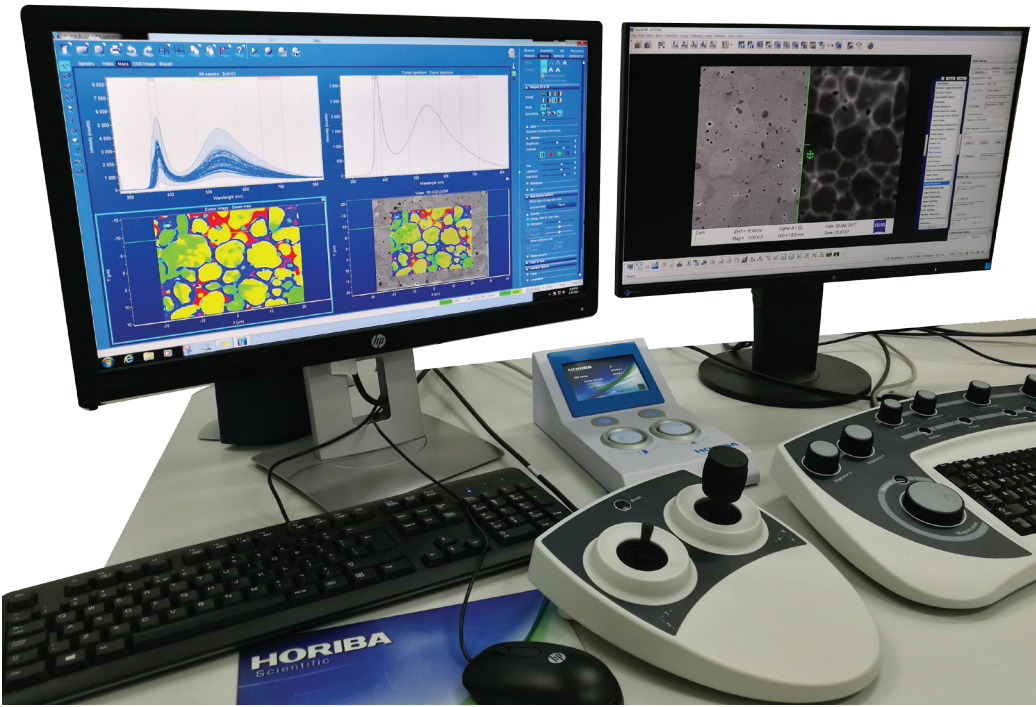
**Application note CL29*



Classical Least Square fitting using multivariate analysis allowing the identification of the substrate and CVD layer region from their emission spectrum*



Luminescence spectra corresponding to the central HPHT substrate region (blue) and to the lateral CVD film region (red). Bright emission from NV centers at 575 nm is detected.*



Al doped ZnO ceramics hyperspectral CL image
Samples courtesy of Materials Genome Institute, Shanghai University, Professor Hui Gu

Find out more at www.horiba.com/cathodoluminescence

	Imaging CL	CL Imaging and Spectroscopy			Raman, CL and PL imaging and spectroscopy
	i-CLUE-e	F-CLUE-e	F-CLUE-p	H-CLUE-p	R-CLUE-p
Spectral range	Panchromatic CL	UV-VIS or VIS-NIR fiber dependent		UV - VIS - NIR Free space achromatic coupling	UV-VIS or VIS-NIR fiber dependent
Diamond turned collection mirror	-ellipsoidal mirror short & long working distances		-parabolic mirror short & long working distances		
200 mm retractable interface	Manual fine adjustment under vacuum		Motorized with fine adjustment under vacuum		
CL imaging detector	Panchromatic CL with ambient PMT	Standard: Panchromatic and monochromatic CL with ambient PMT Options: Cooled PMT, IGA monochannel, photon counting PMT, time resolved PMT, RGB filters			
CL spectrometer type	N/A Option upgrade to F-CLUE-e	microHR (180mm) or iHR 320 (320 mm focal length), up to 3 grating turret, up to 2-entrance / 2 exit		iHR 320, iHR 550 (320 to 550 mm focal length), up to 3 grating turret, 2-entrance/2exit	
Upgrade available	Upgrade to F-CLUE-e	Upgrade from PL, MicOS systems			Upgrade from LabRAM HR Series
Electron beam control	SEM software	CL-LINK for multiple acquisition processing (Analog, pulse mode, SE), Mapping linescan, point measurement, Synchronization with spectroscopic detection, Control by external scan input of Electron Microscope			
Software	SEM software	Spectroscopy and imaging powered by LabSpec 6™			
Remote Controller	Included	Optional			
CLUE Accessories	N/A	Polarization, ND filter, camera, EMCCD etc			
SEM Accessories	N/A	LN2, He cryo-stages, EBIC detector and many other accessories to complement our CLUE Series add-on detectors			

